MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear functions

02

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

B

B. $y = -\frac{1}{10}xx - 4x + 5^2$

Choice B is correct. Each of the given choices is an equation of the form $y = -\frac{1}{10}xx - a^m x + b^n$, where a, b, m, and n are positive constants. In the xy-plane, the graph of an equation of this form has x-intercepts at $x = 0$, $x = a$, and $x = -b$. The graph shown has x-intercepts at $x = 0$, $x = 4$, and $x = -5$. Therefore, $a = 4$ and $b = 5$. Of the given choices, only choices A and B have $a = 4$ and $b = 5$. For an equation in the form $y = -\frac{1}{10}xx - a^m x + b^n$, if all values of x that are less than $-b$ or greater than a correspond to negative y-values, then the sum of all the exponents of the factors on the right-hand side of the equation is even. In the graph shown, all values of x less than -5 or greater than 4 correspond to negative y-values. Therefore, the sum of all the exponents of the factors on the right-hand side of the equation $y = -\frac{1}{10}xx - 4^m x + 5^n$ must be even. For choice A, the sum of these exponents is $1 + 1 + 1$, or 3, which is odd. For choice B, the sum of these exponents is $1 + 1 + 2$, or 4, which is even. Therefore, $y = -\frac{1}{10}xx - 4x + 5^2$ could be the equation of the graph shown.

Choice A is incorrect. For the graph of this equation, all values of x less than -5 correspond to positive, not negative, y-values.

Choice C is incorrect. The graph of this equation has x-intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x-intercepts at $x = -5$, $x = 0$, and $x = 4$.

Choice D is incorrect. The graph of this equation has x-intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x-intercepts at $x = -5$, $x = 0$, and $x = 4$.

Q2

D

D. $f(x) = 3{,}000(0.98)^x$

Choice D is correct. Because the change in the number of students decreases by the same percentage each year, the relationship between the number of students and the number of years can be modeled with a decreasing exponential function in the form $f(x) = a(1 - r)^x$, where $f(x)$ is the number of students, a is the number of students in 2015, r is the rate of decrease each year, and x is the number of years since 2015. It's given that 3,000 students were enrolled in 2015 and that the rate of decrease is predicted to be 2%, or 0.02. Substituting these values into the decreasing exponential function yields $f(x) = 3{,}000(1 - 0.02)^x$, which is equivalent to $f(x) = 3{,}000(0.98)^x$.

Choices A, B, and C are incorrect and may result from conceptual errors when translating the given information into a decreasing exponential function.

Q3

B

B. 55

Choice B is correct. For a quadratic function defined by an equation of the form $gx = ax - h^2 + k$, where a, h, and k are constants and $a > 0$, the minimum value of the function is k. In the given function, $a = 1$, $h = 0$, and $k = 55$. Therefore, the minimum value of the given function is 55.

Choice A is incorrect. This is the value of x for which the given function reaches its minimum value, not the minimum value of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

D

D. 9

Choice D is correct. It’s given that the equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height, in meters, of the object t seconds after it is kicked. It follows that the height, in meters, from which the object was kicked is the value of h when $t = 0$. Substituting 0 for t in the equation $h = -4.9t^2 + 7t + 9$ yields $h = -4.90^2 + 70 + 9$, or $h = 9$. Therefore, the object was kicked from a height of 9 meters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5

B

B. -57

Choice B is correct. For a quadratic function defined by an equation of the form $px = ax - h^2 + k$, where a, h, and k are constants and $a > 0$, the minimum value of the function is k. Subtracting 57 from both sides of the given equation yields $px = x^2 - 57$. This function is in the form $px = ax - h^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

D

D. 270

Choice D is correct. The value of $f0$ is the value of fx when $x = 0$. Substituting 0 for x in the given function yields $f0 = 2700.1^0$, or $f0 = 2701$, which is equivalent to $f0 = 270$. Therefore, the value of $f0$ is 270.

Choice A is incorrect. This is the value of x, not fx.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f1$, not $f0$.

Q7

B

B. The value of the bank account is estimated to be approximately 243 dollars in 1962.

Choice B is correct. It's given that the function $fx = 2061.034^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. It follows that fx represents the estimated value, in dollars, of the bank account x years after 1957. Since the value of $f5$ is the value of fx when $x = 5$, it follows that "$f5$ is approximately equal to 243" means that fx is approximately equal to 243 when $x = 5$. In the given context, this means that the value of the bank account is estimated to be approximately 243 dollars 5 years after 1957. Therefore, the best interpretation of the statement "$f5$ is approximately equal to 243" in this context is the value of the bank account is estimated to be approximately 243 dollars in 1962.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q8

D

D. $fx = x^2 + 5x + 14$

Choice D is correct. The equation of a quadratic function can be written in the form $fx = ax - h^2 + k$, where a, h, and k are constants. It's given in the table that when $x = -1$, the corresponding value of fx is 10. Substituting -1 for x and 10 for fx in the equation $fx = ax - h^2 + k$ gives $10 = a{-1} - h^2 + k$, which is equivalent to $10 = a1 + 2h + h^2 + k$, or $10 = a + 2ah + ah^2 + k$. It's given in the table that when $x = 0$, the corresponding value of fx is 14. Substituting 0 for x and 14 for fx in the equation $fx = ax - h^2 + k$ gives $14 = a0 - h^2 + k$, or $14 = ah^2 + k$. It's given in the table that when $x = 1$, the corresponding value of fx is 20. Substituting 1 for x and 20 for fx in the equation $fx = ax - h^2 + k$ gives $20 = a1 - h^2 + k$, which is equivalent to $20 = a1 - 2h + h^2 + k$, or $20 = a - 2ah + ah^2 + k$. Adding $20 = a - 2ah + ah^2 + k$ to the equation $10 = a + 2ah + ah^2 + k$ gives $30 = 2a + 2ah^2 + 2k$. Dividing both sides of this equation by 2 gives $15 = a + ah^2 + k$. Since $14 = ah^2 + k$, substituting 14 for $ah^2 + k$ into the equation $15 = a + ah^2 + k$ gives $15 = a + 14$. Subtracting 14 from both sides of this equation gives $a = 1$. Substituting 1 for a in the equations $14 = ah^2 + k$ and $20 = ah^2 - 2ah + a + k$ gives $14 = h^2 + k$ and $20 = 1 - 2h + h^2 + k$, respectively. Since $14 = h^2 + k$, substituting 14 for $h^2 + k$ in the equation $20 = 1 - 2h + h^2 + k$ gives $20 = 1 - 2h + 14$, or $20 = 15 - 2h$. Subtracting 15 from both sides of this equation gives $5 = -2h$. Dividing both sides of this equation by -2 gives $-\frac{5}{2} = h$. Substituting $-\frac{5}{2}$ for h into the equation $14 = h^2 + k$ gives $14 = -\frac{5}{2}^2 + k$, or $14 = \frac{25}{4} + k$. Subtracting $\frac{25}{4}$ from both sides of this equation gives $\frac{31}{4} = k$. Substituting 1 for a, $-\frac{5}{2}$ for h, and $\frac{31}{4}$ for k in the equation $fx = ax - h^2 + k$ gives $fx = x + \frac{5}{2}^2 + \frac{31}{4}$, which is equivalent to $fx = x^2 + 5x + \frac{25}{4} + \frac{31}{4}$, or $fx = x^2 + 5x + 14$. Therefore, $fx = x^2 + 5x + 14$ defines f.

Choice A is incorrect. If $fx = 3x^2 + 3x + 14$, then when $x = -1$, the corresponding value of fx is 14, not 10.

Choice B is incorrect. If $fx = 5x^2 + x + 14$, then when $x = -1$, the corresponding value of fx is 18, not 10.

Choice C is incorrect. If $fx = 9x^2 - x + 14$, then when $x = -1$, the corresponding value of fx is 24, not 10, and when $x = 1$, the corresponding value of fx is 22, not 20.

Q9

A

A. $0, 11$

Choice A is correct. The x-coordinate of any y-intercept of a graph is 0. Substituting 0 for x in the given equation yields $g0 = 11\frac{1}{12}^{0}$. Since any nonzero number raised to the 0th power is 1, this gives $g(0) = 11 \cdot 1$, or $g(0) = 11$. The y-intercept of the graph is, therefore, the point $0, 11$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

D

D. The estimated initial population for this species and area in 2008

Choice D is correct. Substituting 0 for x in the given equation yields $f0 = 3{,}0000.75^{0}$, which is equivalent to $f0 = 3{,}0001$, or $f0 = 3{,}000$. It's given that the function estimates the species' population x years after 2008, so it follows that the estimated population of the species is $3{,}000$ in 2008. Therefore, the best interpretation of $3{,}000$ in this context is the estimated initial population for this species and area in 2008.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. The estimated percent decrease in the population for this species and area each year after 2008 is 25%, not $3{,}000$.

Choice C is incorrect. The estimated population for this species and area 8 years after 2008 is $3{,}0000.75^{8}$, or approximately 300, not $3{,}000$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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